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RESEARCH

Video game practice is associated with enhanced value-guided exploitation under probabilistic uncertainty

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Abstract

This study examined whether habitual video game play influences reinforcement learning dynamics, feedback adaptation, consolidation, and motivational biases. Two groups of participants (gamers and controls) completed a Probabilistic Selection Task that assessed learning from positive and negative feedback across three phases: Learning, Test, and Transfer. Mixed-effects modeling revealed that gamers showed enhanced learning trajectories, particularly under high-uncertainty conditions; however, computational modeling indicated no differences in learning rate (α), suggesting comparable value updating across groups. Gamers exhibited a higher tendency toward model-based exploitative (value-consistent) choices during learning compared to controls. In the Test phase, gamers demonstrated higher accuracy, especially on difficult stimulus pairs, suggesting more effective use of learned value representations under no-feedback conditions. This was further supported by greater model-based exploitative choices for challenging pairs. However, while no group differences emerged in transfer-phase approach/avoidance biases (VLBI), gamers showed greater decision consistency (higher inverse temperature, β) and increased exploitative choices in high-value (A-present) trials, indicating more deterministic value-guided behavior during generalization. These findings suggest that habitual video game play improves how efficiently learned values are translated into action under uncertainty, highlighting the potential of game-like environments to enhance value-guided decision-making and adaptive behavior in educational and clinical settings.

Introduction

Video games represent one of the most ubiquitous forms of cognitive engagement in modern life, functioning as ecologically valid environments for reinforcement learning. Unlike static tasks, games require players to continuously make decisions under uncertainty, where feedback is often probabilistic, delayed, or ambiguous (Bavelier & Green, 2019; Sutton & Barto, 2018). To succeed, players must estimate the value of their actions and update their predictions based on whether an outcome yielded a benefit (reward) or a cost (punishment). Consequently, habitual video game play has been associated with enhanced cognitive flexibility and probabilistic learning ability (Bavelier & Green, 2012). However, it remains unclear whether these performance advantages stem from a general improvement in learning or specific alterations in how positive versus negative feedback is processed.

While the principles of reinforcement learning are universal—tracking value to optimize future outcomes—individuals differ significantly in their sensitivity to reinforcement contingencies (Frank et al., 2004; Palminteri et al., 2015). Some learners are primarily driven by "approach" biases (learning to choose the most rewarding option), while others are driven by "avoidance" biases (learning to reject the most punishing option) (Corr, 2004; Kim et al., 2015). Distinguishing between these motivational mechanisms is critical, as they rely on distinct neural substrates within the basal ganglia and predict different behavioral outcomes in educational and clinical settings (Petersburs et al., 2022; Bijttebier et al., 2009).

Previous research has begun to explore how gaming interacts with probabilistic learning. Notably, Schenk and

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colleagues found that video game players outperform non-players on the Weather Prediction Task (WPT), a classic paradigm of probabilistic categorization (Schenk et al., 2017). Although the WPT effectively measures the ability to predict outcomes from complex cues, it does not dissociate reward-driven learning from punishment-driven learning or assess how these value representations are consolidated when feedback is removed. To address this limitation, the Probabilistic Selection Task (PST) offers a complementary framework (Frank et al., 2004). The PST uses a training and transfer design that allows researchers to isolate approach (Choose-A) and avoidance (Avoid-B) biases, providing a more granular view of the underlying learning mechanisms. This paradigm, then, can be used to examine whether habitual video game play is associated with systematic differences in approach- versus avoidance-based learning biases.

However, most prior applications of the PST focus on the final outcome of learning—that is, the consolidated bias observed in the transfer phase (Palminteri et al., 2015; Petersburs et al., 2022). Focusing solely on this endpoint may obscure the dynamic, trial-by-trial adaptations that occur during learning. These acquisition dynamics may be particularly relevant for understanding the cognitive impact of video gaming. Reinforcement learning theory posits that prediction errors—the discrepancy between expected and actual outcomes—drive the updating of value representations (Sutton & Barto, 2018). Given that commercial games frequently demand rapid adaptation to shifting probabilities, habitual gamers may not only possess stronger consolidated biases but also exhibit distinct trajectories of *exploration and exploitation* during learning. A trial-by-trial analysis would make it possible to determine if gamers are more efficient at stabilizing their choices under conditions of high uncertainty (low-contrast probabilistic pairs), a nuance that standard aggregate measures often miss (Gläscher et al., 2010; Niv et al., 2012).

Characterizing these specific reinforcement profiles has implications beyond gaming. If habitual play is associated with increased sensitivity to reward signals—consistent with evidence showing that video game play activates the mesolimbic dopaminergic system (Koepp et al., 1998; Bavelier & Green, 2019)—this could inform the design of instructional feedback in gamified education or therapeutic interventions. For instance, learners with high reward sensitivity may engage more deeply with immediate positive feedback but struggle with delayed reinforcement, whereas avoidance-oriented learners may require different scaffolding strategies (Gan et al., 2021; Bembenutty et al., 2004).

The present study investigated whether habitual video game play is associated with differences in the dynamics and computational signatures of reinforcement learning. Using a modified Probabilistic Selection Task, we examined behavioral performance across Learning, Test, and Transfer phases and complemented these analyses with model-based indices of value updating, choice consistency, and exploration under uncertainty. In this context, exploration and exploitation were defined relative to the computational model's trial-by-trial value estimates rather than the task's true reward probabilities. Specifically, exploitation refers to selecting the option with the highest model-estimated expected value on a given trial, whereas exploration refers to selecting an alternative whose estimated value is lower than that of the currently best option. Under this view, exploration reflects a departure from greedy choice with respect to the model's internal value representation (Daw et al., 2006; Daw & Doya, 2006).

We tested two primary hypotheses. First, based on trial-by-trial computational modeling, we hypothesized that gamers would exhibit more efficient acquisition of probabilistic stimulus values, reflected in faster or more adaptive updating of value estimates and earlier stabilization of exploitative choice policies, particularly under low-contrast conditions where feedback is noisy. Second, we hypothesized that gamers would show a stronger approach-oriented bias—indexed by a preference for the most rewarding stimulus—whereas non-gamers would exhibit a relatively stronger avoidance-oriented bias.

Additionally, we examined whether any learning-related advantages would extend beyond the acquisition phase to the no-feedback test and transfer phases, reflecting more stable value representations and decision policies. By combining mixed-effects behavioral analyses with computational modeling, this study aims to clarify whether gaming experience is associated primarily with enhanced dynamic feedback integration during acquisition or also with more stable motivational biases in generalized decision-making under uncertainty.

Methods and materials

Participants

The present study included a total of 72 young adults: 36 non-players (Control group; 17 males, $M=22.17$, $SD=3.02$; 19 females, $M=21.78$, $SD=2.46$) and 36 video game players (Gamer group; 19 males, $M=21.84$, $SD=2.50$; 17 females; $M=21.67$, $SD=2.56$). Primary game genres for the gamer group are presented in Table S1. Participants were actively enrolled college students who completed a brief clinical history questionnaire; none reported medical, neurological, or psychological conditions that could compromise cognitive performance. Recruitment materials stated

that compensation was performance-contingent, with the three highest-scoring participants earning gift cards worth $800, $500, and $300 MXN. This incentive structure was designed to ensure sustained attention and optimize task engagement. To mitigate social pressure during the task, participants were informed that the final scoreboard (using anonymized IDs) would only be disclosed after the entire study was concluded.

The Gamer group was required to actively play video games for seven or more hours per week ($M = 20.55$ h/week, $SD = 11.2$), maintained for at least six months prior to the assessment. Following common practice in the field, frequent players are typically defined as those engaging in ≥5 h/week of video game play for ≥6–12 months (Green et al., 2012); we used a slightly stricter cutoff (≥7 h/week for ≥6 months) to maximize group separation while remaining within the canonical range. In contrast, the Control group was restricted to participants reporting between zero and three hours of weekly gameplay ($M = 1.38$ h/week, $SD = 1.15$). To ensure group separation, we excluded participants who reported a regular gaming habit at any prior life stage or whose current gameplay (even within the 0–3 h range) had been sustained for more than one month at the time of assessment. All participants had normal or corrected-to-normal vision and no history of psychological, psychiatric, or neurological conditions that could interfere with the study.

To characterize the sample and screen for variables that might influence the results, we measured impulsivity (Barratt Impulsiveness Scale, BIS-11) and sensitivity to punishment and reward (The Sensitivity to Punishment and Sensitivity to Reward Questionnaire, SPSRQ). All participants provided written informed consent. The research protocol was reviewed and approved by the Ethics Committee of the Facultad de Ciencias Administrativas y Sociales of the Universidad Autónoma de Baja California (Approval number: 430/2025-1).

Stimuli and task

To assess reinforcement-based learning, the Probabilistic Selection Task (PST) proposed by Frank et al. (2004) was adapted and implemented using PsychoPy® (version 2024.2.1), an open-source software for behavioral experiments (Peirce et al., 2019). This task evaluates whether individuals tend to learn better through positive feedback/reward (approach learning) or negative feedback/punishment (avoidance learning). To maintain attention, stimuli were contextualized through a narrative frame. Participants were told they were deciphering pairwise combinations of symbols from an invented ancient language, referred to as "The Codebreaker's Challenge." Task structure and contingencies remained identical to the standard PST. The task consisted of three phases: Learning, Test, and Transfer. (The exact instructions for each phase are provided in the Supplementary Material, S2). Total task duration ranged from 15 to 60 min, depending on the participant.

In the Learning phase, participants were shown one of three symbol pairs (A/B, C/D, E/F) for 3500 ms. The left–right spatial positions of the two symbols were randomized on every trial. Each symbol was associated with a specific reinforcement probability (Fig. 1). In the A/B pair, choosing symbol A yielded positive feedback on 80% of trials and negative feedback on 20%, whereas choosing symbol B resulted in the opposite probabilities (20% positive, 80% negative). Similarly, the C/D and E/F pairs were associated with reinforcement contingencies of 70/30 and 60/40, respectively (Fig. 1). After a response (left or right key press), the selected symbol was highlighted in blue for 300 ms, followed by positive or negative feedback displayed for 500 ms. Participants were shown their current score in the top-right corner, the number of remaining attempts in the top-left corner, and whether they gained or lost points in the center of the screen. Correct responses earned ten points, while incorrect ones resulted in a ten-point deduction. Participants were instructed to use feedback to learn which symbol to choose in order to maximize their score; however, they were not informed that each stimulus pair had associated reinforcement probabilities. If the response time exceeded 3500 ms, a message appeared on the screen prompting the participant to respond faster, and the trial was not scored. Each stimulus pair was presented 20 times in random order, for a total of 60 trials (Frank et al., 2004; Peterburs et al., 2022).

After completing the Learning phase, participants proceeded to the Test phase. The task structure remained the same, except that no feedback was provided, requiring participants to rely on previously learned stimulus–outcome associations. Each stimulus pair was presented 10 times in random order, resulting in 30 trials. The Transfer phase began once participants achieved at least 80% accuracy in the Test phase—that is, 24 correct responses out of 30. If this criterion was not met, the Learning and Test phases were repeated after an optional one-minute break. Each participant could repeat these phases up to five times; if the accuracy threshold was still not reached, the Transfer phase proceeded regardless (Frank et al., 2004; Peterburs et al., 2022).

In the Transfer phase, the original stimulus pairs were recombined into novel pairings. The symbol with the highest probability of positive feedback ("A") was paired with all symbols it had not previously been paired with, resulting in the combinations A/C, A/D, A/E, and A/F. Similarly,

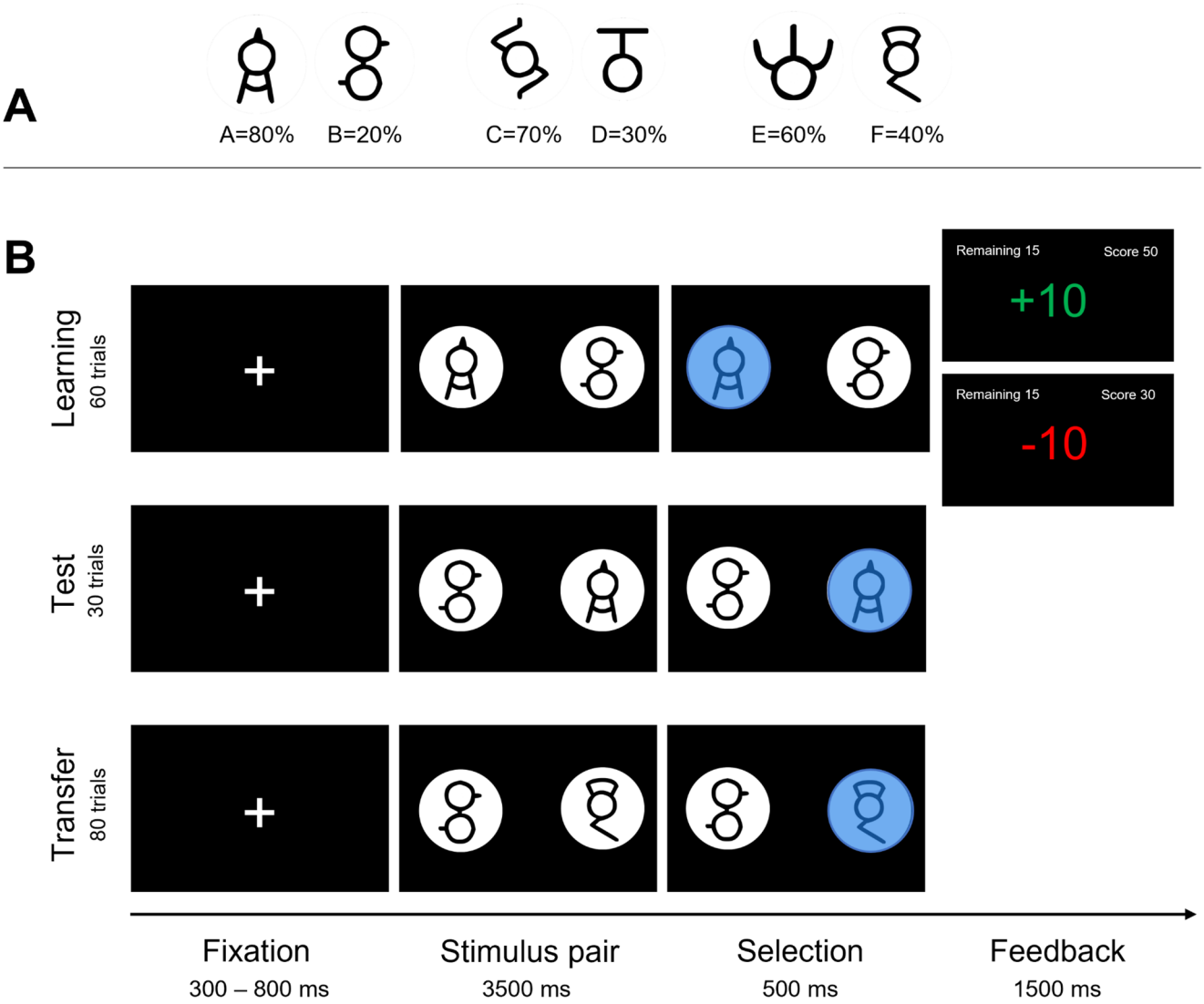


Fig. 1 Structure of the Probabilistic Selection Task. Note. (**A**) Stimulus pairs and their programmed reinforcement probabilities. Each abstract symbol was associated with a fixed probability of receiving positive feedback: symbol A (80% reward) paired with B (20%), C (70%) with D (30%), and E (60%) with F (40%). Participants were not informed of these contingencies and inferred them through trial-by-trial feedback. (**B**) Example trial sequence for the three task phases—Learning, Test, and Transfer. Each trial began with a fixation screen (300–800 ms), followed by the presentation of one stimulus pair (3500 ms). Participants selected one of the two symbols by pressing a key, after which the chosen symbol was highlighted (500 ms). During the Learning phase, feedback was displayed for 1500 ms, indicating point gain (+10) or loss (–10), along with the running score and remaining trials. No feedback was provided during the Test or Transfer phases. In the Transfer phase, original stimulus pairs were recombined to form novel pairs involving A or B with previously unseen partners, allowing assessment of approach (choosing A) and avoidance (avoiding B) tendencies based on learned value representations

the symbol with the lowest probability of positive feedback ("B") was paired with the remaining symbols (B/C, B/D, B/E, and B/F). Left–right positions continued to be randomized on every trial, following the same procedure used during the Learning and Test phases. The Transfer phase consisted of 80 trials in total, with symbols A and B appearing in 40 trials each. No feedback was provided, and the stimulus sequence and timing were identical to those in the Test phase (Frank et al., 2004; Peterburs et al., 2022).

Data processing

All statistical analyses were conducted in R (version 4.3.1). Linear and generalized linear mixed-effects models were fitted using the lme4 package. When required, mixed-effects models were optimized using the bobyqa optimizer. Statistical inference was based on Wald tests for fixed effects in mixed models and Welch's t-tests for between-group comparisons, where appropriate.

Participants were excluded if more than 20% of trials were missing; however, no participant met this criterion (maximum = 6.6%). Reaction time (RT) distributions were screened for within-subject outliers: trials with absolute z-scores greater than 2.5 were treated as missing to prevent extreme values from biasing subsequent analyses. This led to the exclusion of 1.6% of trials. No additional trimming was applied.

Behavioral analysis

Learning phase analysis

First, the number of Learning–Test cycles required to reach the Transfer phase was analyzed using a negative binomial regression model with Group as the predictor. This approach was selected because the outcome variable represented a discrete count rather than a continuous measure.

Performance across the Learning phase was analyzed by binning trials into blocks of ten. Mean accuracy per block was calculated separately for each participant and stimulus pair (A/B, C/D, E/F). A generalized linear mixed-effects model (GLMM) was then fitted to these data, including fixed effects for Group (Control vs. Gamer), Block (linear term), Pair (A/B, C/D, E/F), and all their interactions. Participant-specific random intercepts and random slopes for Block were included to account for variability in baseline performance and learning rates.

Test phase analysis

Given the absence of feedback during the Test phase, learned preferences were assessed by coding accuracy as the selection of the higher-value symbol within each pair (A in A/B, C in C/D, E in E/F). Trials without a recorded response were omitted. Accuracy data were analyzed using logistic mixed-effects regression with fixed effects for Group, Pair, and their interaction, and participant-specific random intercepts.

Transfer phase analysis

To quantify motivational bias, we computed the Valence Learning Bias Index (VLBI) for each participant as the difference between the proportion of A selections and the proportion of B avoidances. Positive VLBI values indicate a bias toward reward-driven approach, while negative values indicate a bias toward punishment-driven avoidance. Group differences in VLBI were analyzed using a simple linear regression.

Computational modelling

To complement the behavioral analyses, we fitted a subject-specific reinforcement-learning model to trial-by-trial choices during the Learning phase and then used the learned value representations to characterize performance during the subsequent no-feedback phases. The goal of this analysis was not to replace the behavioral models but to decompose performance into separable computational components: (a) value updating during feedback-based acquisition, (b) the extent to which learned values guide choices in the Learning, Test, and Transfer phases, and (c) the degree of exploratory versus exploitative choice under different levels of uncertainty. Trials without a recorded response were excluded, consistent with the behavioral analyses.

On each Learning trial t, participants chose between two stimuli, i and j, from the set $S = \{A, B, C, D, E, F\}$. Each stimulus $s \in S$ was associated with a latent action value $Q_{s,t}$. In the fitted implementation, outcomes were coded as $r_t \in \{-1, +1\}$, where positive feedback was coded as +1 and negative feedback as -1. Initial values were set to zero for all stimuli.

Choice probabilities were generated using a softmax decision rule, expressed equivalently as a logistic function of the value difference between the left and right options:

$$P_{t(choose\ left)} = \frac{exp\,(\beta\, Q_{L,t})}{(exp\,(\beta\, Q_{L,t}) + exp(\beta\, Q_{R,t})\,)} = \frac{1}{(1 + exp(-\beta\,(Q_{L,t} - Q_{R,t}))\,)}$$

where $\beta > 0$ is the inverse temperature parameter. Larger values of β indicate more deterministic, exploitative choice, whereas smaller values indicate more stochastic behavior.

After each choice, only the chosen option was updated according to a Rescorla-Wagner learning rule. The prediction error on trial t was defined as:

$$\delta_t = r_t - Q_{c_t,t}$$

and the chosen value was updated as:

$$Q_{c_t,t+1} = Q_{c_t,t} + \alpha\, \delta_t$$

where c_t denotes the chosen stimulus and $\alpha \in (0,1)$ is the learning-rate parameter. The value of the unchosen option remained unchanged.

For each participant, the Learning-phase parameters α and β were estimated by maximizing the log-likelihood of the observed sequence of choices across all available learning runs. The negative log-likelihood was defined as:

$$log\ L\ (\alpha, \beta) = \sum_{t=1}^{T} log\ P_t\ (c_t \mid Q_t, \alpha, \beta)$$

and the fitted objective function was the corresponding negative log-likelihood:

$$L(\alpha, \beta) = -\sum_{t=1}^{T} log\ P_t\ (c_t \mid Q_t, \alpha, \beta)$$

Optimization was performed using the L-BFGS-B algorithm with multiple starting values to reduce sensitivity to local minima. Parameter bounds were constrained to $0.001 \leq \alpha \leq 0.999$ and $0.01 \leq \beta \leq 500$. Because α is bounded between 0 and 1, group comparisons were conducted on its logit-transformed values, whereas β was analyzed on the log scale to reduce skewness and improve distributional properties for linear modeling.

Using each participant's fitted Learning-phase parameters, we reconstructed trial-by-trial latent variables during acquisition. For every trial, we extracted the model-implied probability of the observed choice, P_t (c_t), as well as the current values of the left and right stimuli. We also derived a model-based exploration indicator: choices were classified as exploitative when the participant selected the option with the higher current Q-value and exploratory when the lower-valued option was selected.

Trials in which $Q_{(i,\ t)} = Q_{(j,\ t)}$ were treated as indeterminate and excluded from the exploitative versus exploratory classification. For inferential analyses during the Learning phase, the primary focus was restricted to the first Learning run, which was available for all participants and therefore provided the most comparable estimate of initial feedback-based acquisition. This restriction was adopted because later Learning runs occurred only when participants had not yet met the criterion in the preceding Test phase and thus did not represent an equivalent stage of acquisition across subjects.

Subject-level group differences in learning rate (α) and inverse temperature (β) were tested using linear models on logit-transformed α and log-transformed β, respectively. To examine how exploitative responding evolved during acquisition as a function of uncertainty, trial-level exploitative choices from the first Learning run were analyzed using a binomial generalized linear mixed-effects model with Group, Trial (1–60), Pair (A/B, C/D, E/F) included as fixed effects, along with all interactions, and participant-specific random intercepts.

The same learned value representations were then carried forward to the Test and Transfer phases. As these phases were performed without feedback, no additional learning-rate parameter was estimated outside the Learning phase. Instead, latent values were treated as fixed summaries of what each participant had already learned, and only phase-specific inverse-temperature parameters were estimated. This separation was intentional: once feedback is absent, the relevant question is no longer how quickly participants update values, but how consistently they express previously learned values in their choices.

End-of-run Q-values were computed by carrying forward Rescorla-Wagner updates across all Learning runs completed by a given participant, rather than resetting values at the beginning of each repeated run. This approach preserved each participant's cumulative learning history and matched the task structure, in which repeated Learning-Test cycles continued until the performance criterion was reached. In other words, later runs were treated as additional opportunities to refine the same value representations rather than as independent learning episodes.

For the Test phase, a separate β_{test} parameter was estimated for each participant and Test run using the Q-values obtained at the end of the corresponding Learning run. Because no feedback was delivered, no value updating occurred and the likelihood depended only on the fixed learned values and the observed choices. Analyses focused on the first Test run, which was available for all participants and therefore yielded the most comparable between-subject dataset. Group differences in β_{test} were assessed using linear models on log-transformed values. In addition, trial-level exploitative choices from the first Test run were analyzed using a binomial generalized linear mixed-effects model with Group, stimulus Pair, and their interaction as fixed effects and participant-specific random intercepts. This analysis was designed to test whether group differences in value-guided choices during the Test phase varied across the learned stimulus pairs, particularly as a function of pair difficulty, rather than as a function of within-phase temporal progression.

For the Transfer phase, a single $\beta_{transfer}$ parameter was estimated for each participant using the Q-values carried forward from the end of the final Learning run. As in the Test phase, no value updating occurred and the likelihood depended only on the fixed learned values and the observed choices. Group differences in $\beta_{transfer}$ were assessed using linear models on log-transformed values. Trial-level exploitative and exploratory choices were also derived for descriptive and inferential purposes. Exploitative choices were analyzed using a binomial generalized linear mixed-effects model with Group, Trial Type (choose-A versus avoid-B), and their interaction as fixed effects and participant-specific random intercepts.

Results

Sample characteristics

Table 1 summarizes the means and standard deviations for weekly gaming hours, the Barratt Impulsiveness Scale

Table 1 Sample characterization

Variable	Control M (SD)	Gamer M (SD)	t	*p*	Cohen´s d
Weekly Gaming Hours	1.83 (1.87)	20.56 (11.05)	-9.88	**<0.001****	-2.32
BIS-11	62.47 (6.66)	59.94 (11.73)	1.11	0.27	0.26
SPSRQ Reward	11.50 (5.17)	10.61 (4.34)	1.40	0.16	0.32
SPSRQ Punishment	12.81 (5.26)	12.77 (4.82)	0.02	0.98	0.005

BIS-11 = Barratt Impulsiveness Scale; *SPSRQ* the sensitivity to punishment and sensitivity to reward questionnaire. Significant comparisons are presented in bold * $p \leq .05$, ** $p < .001$

(BIS-11), and the Sensitivity to Punishment and Sensitivity to Reward Questionnaire (SPSRQ), providing an overview of the sample. No group differences were observed in any measure, except for weekly gaming hours in the last six months.

Learning phase

A negative binomial regression model which analyzed Learning-Test cycles revealed no significant effect of Group (β = -0.184, SE = 0.127, z = -1.45, p = .147), indicating that the Gamer and Control groups did not differ in the number of cycles required to progress to the Transfer phase. Although the model-estimated number of cycles was numerically lower in the Gamer group (M = 3.17, 95% CI [2.64, 3.80]) than in the Control group (M = 3.81, 95% CI [3.22, 4.50]), this difference was not statistically significant.

The generalized linear mixed-effects model (Fig. 2; Table 2) examining accuracy across the Learning phase revealed significant main effects of Block (z = 5.66, p < .001), indicating overall improved performance over time, and Pair, specifically highlighting lower accuracy for pair E/F compared to A/B (z = -4.38, p < .001). A significant three-way interaction among Group, Block, and Pair was also observed, with gamers showing differential learning trajectories compared to controls, particularly for pairs C/D (z = -2.03, p = .043) and E/F (z = -2.66, p = .008). This suggests that while both groups improved over blocks, Gamers exhibited distinct learning patterns depending on stimulus difficulty.

Table 2 Model for accuracy across learning blocks

Effect	β	SE	Z	*p*
Intercept	0.333	0.113	2.933	0.003
GroupGamer	0.103	0.164	0.632	0.527
Block	**0.060**	**0.010**	**5.662**	**<0.001**
PairC/D	-0.113	0.110	-1.030	0.302
PairE/F	**-0.476**	**0.108**	**-4.379**	**<0.001**
GroupGamer: Block	0.017	0.015	1.148	0.250
GroupGamer: PairC/D	-0.013	0.161	-0.084	0.932
GroupGamer: PairE/F	**0.470**	**0.160**	**2.931**	**0.003**
Block: PairC/D	**-0.015**	**0.007**	**-2.184**	**0.028**
Block: PairE/F	**-0.029**	**0.007**	**-4.062**	**<0.001**
GroupGamer: Block: PairC/D	**-0.022**	**0.011**	**-2.027**	**0.042**
GroupGamer: Block: PairE/F	**-0.029**	**0.011**	**-2.663**	**0.007**

Bold values indicate statistical significance

The linear regression analysis examining the learning rate parameter (α) during the Learning phase revealed no significant effect of Group (β = 0.053, SE = 0.037, t = 1.44, p = .154), indicating that the groups did not differ in their overall rate of value updating. This model-based result suggests that group differences during learning were not driven by differences in value updating rate.

Similarly, the linear regression analysis of the inverse temperature parameter (β) during the Learning phase showed no significant effect of Group (β = -0.438, SE = 0.287, t = -1.53, p = .131), indicating comparable levels of choice consistency across groups during acquisition. Thus, this result suggests that group differences during learning were not driven by overall differences in choice determinism.

Finally, the mixed-effects logistic regression analysis of exploitative choices during the Learning phase revealed

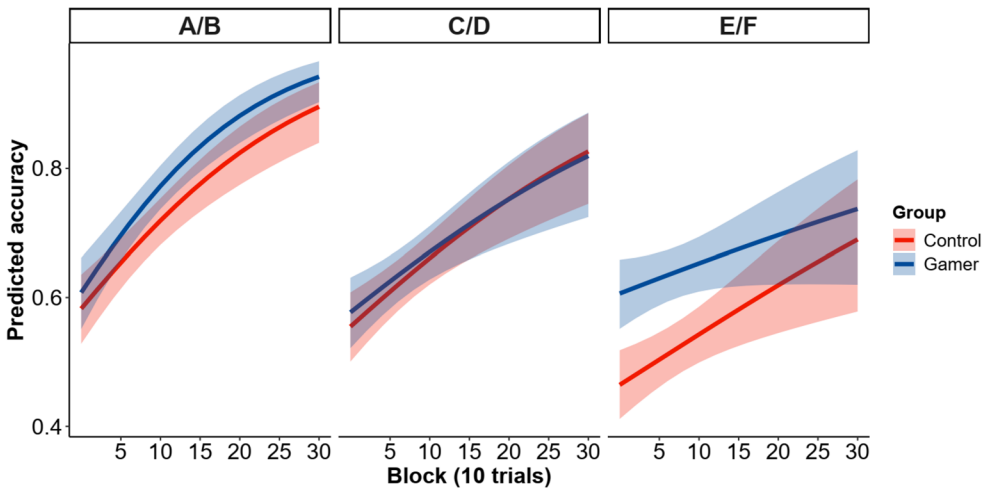


Fig. 2 Learning curves across probabilistic reinforcement conditions. *Note*. Model-based predicted accuracy across successive training blocks (each block = 10 trials) for the three stimulus pairs with different reinforcement contingencies: A/B (80/20), C/D (70/30), and E/F (60/40). Accuracy reflects the probability of selecting the higher-value symbol in each pair. Curves are shown separately for the Control (red) and Gamer (blue) groups. Shaded areas represent 95% confidence intervals around the model estimates

a significant effect of Group (p = .043), with the Control group showing a lower overall probability of model-based exploitative choices and the Gamer group showing a higher tendency to exploit learned value differences (Fig. 3). In addition, a significant effect of Pair was observed (p < .001; see Table 3 for full model statistics), indicating that exploitative choices varied across stimulus pair as a function of reinforcement contingency. No other effects reached significance, suggesting that these group and pair effects reflected overall shifts in exploitative choice levels rather than differential changes across trials.

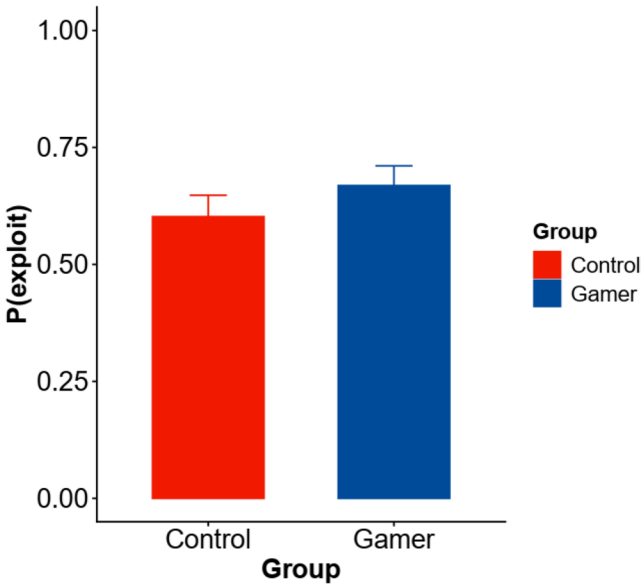


Fig. 3 Model-based exploitative choices during the learning phase by group. *Note*. Mean probability of exploitative choices—selection of the higher estimated value according to the computational model—shown separately for the Control (red) and Gamer (blue) groups. Error bars represent standard errors of the mean

Table 3 Fixed effects from the mixed-effects logistic regression model predicting model-based exploitative choices during the learning phase

Effect	β	SE	Z	p
Intercept	0.558	0.071	7.873	< .001
Group	-0.144	0.071	-2.028	0.043
Trial	-0.004	0.034	-0.122	0.903
PairA/B	0.182	0.049	3.747	< .001
PairC/D	-0.075	0.048	-1.567	0.117
Group × Trial	0.002	0.034	0.048	0.962
Group × PairA/B	-0.005	0.049	-0.094	0.925
Group × PairC/D	0.006	0.048	0.129	0.897
Trial × PairA/B	0.033	0.049	0.67	0.503
Trial × PairC/D	0.023	0.048	0.469	0.639
Group × Trial × PairA/B	-0.034	0.049	-0.699	0.485
Group × Trial × PairC/D	0.041	0.048	0.837	0.402

Bold values indicate statistical significance

Test phase

The mixed-effects logistic regression analysis of accuracy during the Test phase (Fig. 5) revealed significant main effects for Group (z = 1.99, p = .047), indicating that Gamers had higher overall accuracy than Controls, and Pair, with pairs C/D (z = -4.47, p < .001) and E/F (z = -12.58, p < .001) showing significantly lower accuracy compared to pair A/B. The interaction between Group and Pair was significant for pair E/F (z = 2.80, p = .005), but not for the C/D pair (z = 1.09, p = .27), suggesting Gamers performed better, specifically on the most challenging stimulus pair (E/F) compared to Controls (Fig. 4).

At the computational-model level, the linear regression analysis of the inverse temperature parameter (β) showed no significant effect of Group (β = -0.629, SE = 0.411, t = -1.53, p = .130), indicating comparable levels of choice consistency across groups when previously learned values had to be expressed in the absence of feedback.

Furthermore, the mixed-effects logistic regression analysis of exploitative choices during the Test phase revealed a significant effect of Group (p = .014) and a significant Group × Pair interaction (p = .048; see Fig. 5; Table 4 for full model statistics). Post hoc comparisons indicated that the Gamer group showed a significantly higher probability of model-based exploitative choices than the Control group for the C/D pair (OR = 0.599, z = -2.109, p = .035) and the E/F pair (OR = 0.467, z = -3.139, p = .002), whereas no significant group difference was observed for the A/B pair (OR = 0.810, z = -0.968, p = .336). Thus, this model-based result indicates that group differences in exploitative choices during the Test phase were not generalized across all stimulus pairs but were instead driven by the more difficult

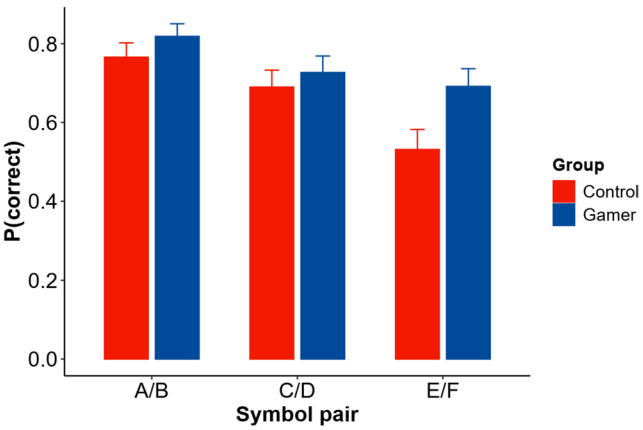


Fig. 4 Accuracy during the test phase across stimulus pairs. *Note*. Mean probability of selecting the higher-value symbol for each learned pair—A/B (80/20), C/D (70/30), and E/F (60/40)—shown separately for the Control (red) and Gamer (blue) groups. Error bars represent standard errors of the mean

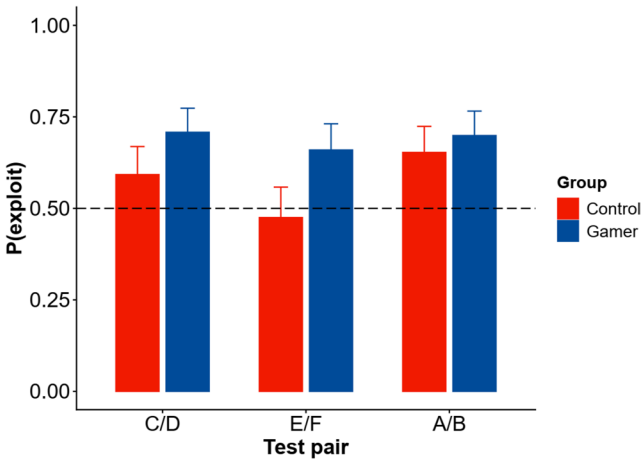


Fig. 5 Model-based exploitative choices during the test phase by group and stimulus pair. *Note*. Mean probability of exploitative choices—selection of the higher estimated value according to the computational model—shown separately for the Control (red) and Gamer (blue) groups across the three test pairs (C/D, E/F, and A/B). The dashed horizontal line indicates chance level (0.50). Error bars represent standard errors of the mean

Table 4 Fixed Effects from the mixed-effects logistic regression model predicting model-based exploitative choices during the test phase

Predictor	β	SE	Z	*p*
Group	-0.247	0.101	-2.45	0.014
Intercept	0.548	0.101	5.412	<0.001
PairA/B	0.08	0.067	1.195	0.232
PairC/B	**-0.268**	**0.068**	**-3.975**	**<0.001**
Group × PairA/B	-0.008	0.067	-0.125	0.9
Group × PairC/D	**-0.133**	**0.067**	**-1.977**	**0.048**

Bold values indicate statistical significance

learned associations, with gamers showing greater value-guided consistency particularly for the C/D and E/F pairs.

Transfer phase

In the Transfer phase, the VLBI, measuring preference toward approach versus avoidance behavior, did not differ significantly between groups (t(70) = -0.26, p = .798).

At the computational-model level, the linear regression analysis of the inverse temperature parameter (β) during the Transfer phase showed a significant effect of Group (β = -0.719, SE=0.318, t = -2.26, p = .027), indicating greater choice consistency in the Gamer group than in the Control group during value generalization (Fig. 6). This suggests that, during the Transfer phase, gamers expressed learned value representations more deterministically when selecting between novel stimulus combinations.

The mixed-effects logistic regression analysis of exploitative choices during the Transfer phase revealed no significant main effect of Group (β = -0.107, SE=0.090, z = -1.19, p = .233), but showed a significant effect of TrialType

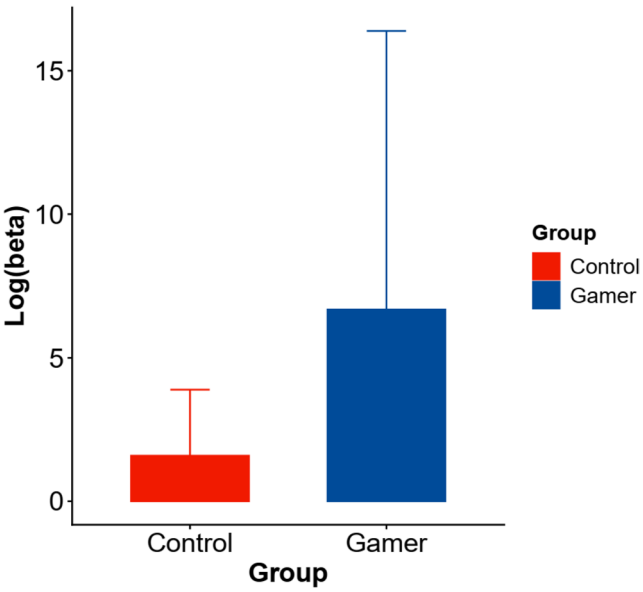


Fig. 6 Inverse Temperature Parameter (B) during the Transfer Phase by Group. *Note.* Mean log-transformed β values derived from the computational model are shown separately for the Control (red) and Gamer (blue) groups. Higher β values indicate more deterministic, value-consistent choices during transfer. Error bars represent standard errors of the mean

(β=0.213, SE=0.030, z=7.12, p < .001) as well as a significant Group × TrialType interaction (β = -0.082, SE=0.030, z = -2.73, p = .006). Post hoc comparisons indicated that the Gamer group showed a significantly higher probability of model-based exploitative choices than the Control group in A-present trials (OR=0.685, df=Inf, z = -1.98, p = .048), whereas no significant group difference was observed in B-present trials (OR=0.951, df=Inf, z = -0.27, p = .788). Thus, this result suggests that group differences during the Transfer phase were specific to trials involving the most positively valued stimulus, with gamers showing greater value-consistent exploitation when stimulus A was present, but not during trials involving stimulus B (Fig. 7).

Discussion

The present study investigated how habitual video game play relates to reinforcement learning dynamics, emphasizing trial-by-trial adaptation across probabilistic contingencies and the subsequent expression of learned value representations during consolidation and transfer in the absence of feedback.

Participants in the Gamer group reported significantly more weekly gaming hours than those in the Control group, with a large effect size, confirming the robustness of the group classification. No significant groups differences were found in impulsivity (BIS-11) or sensitivity to reward and

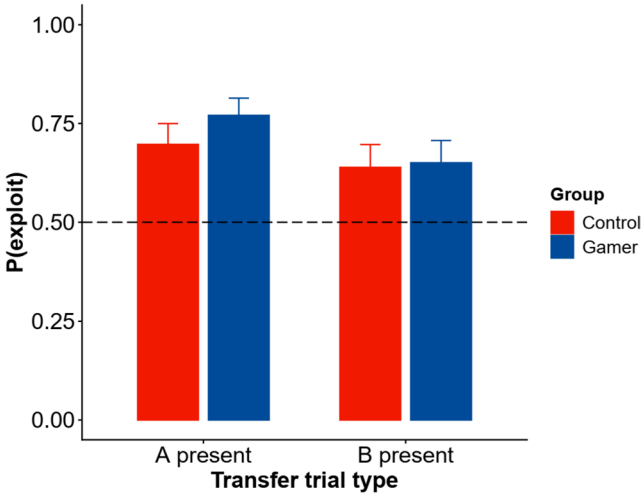


Fig. 7 Model-based exploitative choices during the transfer phase by group and trial type. *Note*. Mean probability of exploitative choices—selection of the higher estimated value according to the computational model—shown separately for the Control (red) and Gamer (blue) groups in A-present and B-present transfer trials. The dashed horizontal line indicates chance level (0.50). Error bars represent standard errors of the mean

punishment (SPSRQ), variables that have been linked to reinforcement learning performance (Raio et al., 2020; Unger et al., 2012). These findings indicate that the groups were comparable in their general self-perceived impulsivity and reinforcement sensitivity, suggesting that observed differences in task performance are unlikely to reflect these dispositional factors.

Based on previous findings linking gaming experience to probabilistic inference (Bavelier et al., 2012) and probabilistic categorization learning outcomes (Schenk et al., 2017), we examined how individuals update value expectations from feedback using trial-by-trial modeling. During the Learning phase, gamers demonstrated superior performance at the behavioral level, as reflected in a significant three-way interaction among Group, Block, and Pair. This interaction indicates that gamers displayed distinct learning trajectories, particularly for the more difficult discrimination pairs (C/D and E/F), suggesting an enhanced capacity to integrate probabilistic feedback under conditions of greater uncertainty.

This interpretation is consistent with the "learning to learn" framework, which proposes that video game experience enhances the rate at which individuals extract task-relevant structure from novel environments (Zhang et al., 2021), as well as with broader evidence indicating improved inference and decision-making efficiency under probabilistic uncertainty in gamers (Bavelier et al., 2012; Green & Bavelier, 2019).

However, at the computational level, no group differences were observed in the learning rate parameter (α), indicating that these performance advantages were not driven by faster value updating. Instead, they appear to reflect differences in how learned values are used to guide behavior during acquisition. Similarly, no group differences were observed in the inverse temperature parameter (β) during the Learning phase, indicating that global choice stochasticity did not differ between groups. Importantly, β provides subject-level index of overall choice determinism, whereas the model-based classification of exploitative choices captures trial-by-trial alignment between behavior and current value estimates. The dissociation between these measures suggests that the increased exploitative choices observed in gamers reflects a more precise or context-sensitive mapping between value estimates and action selection, rather than a generalized increase in deterministic choices. This pattern contrasts with accounts in which enhanced reinforcement learning is primarily reflected in changes to core learning parameters (e.g., higher learning rates) or global decision consistency. Instead, it points to differences in how learned values are selectively expressed in behavior under uncertainty. This interpretation is consistent with evidence suggesting that gaming experience enhances performance in probabilistic and noisy environments by improving the efficiency with which task-relevant information is extracted and used to guide decisions, rather than by uniformly increasing learning rates or general response consistency (Bavelier et al., 2012; Green & Bavelier, 2019).

The model-based exploration–exploitation analysis further supports this pattern. When defined relative to trial-by-trial value estimates, gamers exhibited a higher overall probability of exploitative choices during the Learning phase, indicating a greater tendency to select the option with the highest estimated value. Exploitative choices varied as a function of reinforcement contingency, indicating that both groups adjusted their behavior to the probabilistic structure of the task, with stronger value-consistent choices emerging in conditions with clearer reward contingencies. Notably, this effect emerged independently of learning rate differences, suggesting that gamers did not learn faster, but instead showed more consistent alignment between their choices and their internal value representations. This pattern is consistent with a more efficient stabilization of value-guided decision policies, in which behavior reflects the mapping between learned values and action selection (Sutton & Barto, 2018; Daw & Doya, 2006). Given that video game environments require rapid evidence accumulation, continuous outcome monitoring, and adaptive strategy selection under uncertainty (Green & Bavelier, 2019), these findings suggest that gaming experience may strengthen the mapping between learned values and action selection rather than the updating process itself. No group-by-trial interaction was observed, indicating that the observed differences reflect a stable increase in value-consistent choices across

the Learning phase rather than a faster transition from exploration to exploitation.

Prediction-error–driven learning has been closely linked to dopaminergic signaling, with phasic midbrain dopamine activity proposed to encode reward-prediction errors that guide synaptic modification (Schultz et al., 1997; Daw & Doya, 2006). Although the present study did not include neural measures and revealed no group differences in learning rate parameters, the observed differences in value-guided choice behavior are consistent with the possibility that gaming experience influences how learned value representations are expressed in action selection. Repeated engagement with reward-contingent and feedback-rich environments may therefore enhance the efficiency of decision policies supported by mesolimbic–cortical circuits, rather than altering the core updating mechanisms (Koepp et al., 1998; Green & Bavelier, 2019).

Taken together, these parallels suggest that video games may serve as a naturalistic reinforcement-learning context, in which behavioral patterns reflect adaptations to the specific cognitive demands of gaming—particularly those involving feedback integration and decision-making under uncertainty.

In the Test phase, where participants applied previously learned contingencies without feedback, gamers outperformed controls in overall accuracy, particularly on the most difficult pair (E/F). A significant Group × Pair interaction indicated that gamers retained and applied learned associations more effectively under high ambiguity. Importantly, model-based analyses further supported this interpretation, revealing that gamers exhibited a higher probability of exploitative, value-consistent choices during the Test phase, particularly for the more difficult stimulus pairs (C/D and E/F). The greater tendency toward model-based exploitative choices observed in gamers, particularly for the most difficult pair (E/F), reflects a more efficient use of learned reward contingencies rather than reduced flexibility or behavioral rigidity. Importantly, this pattern does not indicate more globally deterministic behavior, but rather a more consistent alignment between choices and learned value representations under conditions of high uncertainty. This interpretation is consistent with prior findings showing that gamers outperform non-gamers in cognitive flexibility and decision-making under uncertainty (Buelow et al., 2015; Martínez-Maldonado, 2025).

However, no group differences were observed in the inverse temperature parameter (β), suggesting that these effects were not driven by global differences in choice consistency but rather by more precise or selectively expressed value representations under challenging conditions. This pattern extends previous findings from probabilistic learning tasks (Schenk et al., 2017) by suggesting that gaming experience may facilitate the effective deployment of learned value representations when feedback is no longer available.

Such reinforcement-related advantages may also have meaningful implications for learning and adaptation in everyday contexts. Educational environments frequently rely on incremental skill acquisition shaped by corrective feedback, a structure that closely parallels core reinforcement learning mechanisms such as value-guided behavior, sensitivity to learned value contingencies, and feedback integration. Game-like elements—such as immediate reinforcement, adaptive difficulty, and variable reward schedules—can strengthen these processes by aligning motivation with performance demands. For example, educational games that integrate task goals within feedback-rich gameplay have been found to foster both learning and motivation (Habgood et al., 2011). Similarly, a meta-analysis by Wouters et al. (2013) found that serious games outperform conventional instruction, particularly when they incorporate real-time feedback and adaptive challenge. These findings suggest that game-based dynamics amplify reinforcement-driven learning by providing a structured yet flexible framework for monitoring performance, adjusting strategies, and internalizing contingencies. In line with this, feedback and self-regulation have been positively linked to academic success (Bembenutty & Karabenick, 2004), and more recent work indicates that gamified feedback systems can further optimize learners' cognitive engagement as well.

Probabilistic feedback integration and reward sensitivity under uncertainty are directly relevant to the design of educational games. Games that use variable reward schedules, partially reliable feedback, or uncertainty-based progression can encourage learners to update their predictions more effectively and sustain engagement. For instance, an educational puzzle game that delivers hints with controlled probabilistic accuracy (e.g., 70–80%) requires players to infer correct strategies from noisy outcomes, thereby reinforcing the same computational processes—prediction updating and sensitivity to reward contingencies—that support real-world learning. Building on this foundation, our study examined whether habitual video game play is associated with measurable differences in reinforcement learning processes—such as sensitivity to reward and punishment contingencies—that could underlie such transfer.

In the Transfer phase, previously learned pairs were recombined to assess value generalization in the absence of feedback. While no group differences were observed in the Valence Learning Bias Index (VLBI), model-based analyses revealed a more nuanced pattern. Specifically, gamers exhibited significantly higher inverse temperature (β) values, indicating more deterministic, value-consistent responding during transfer. In addition, group differences in exploitative choices emerged selectively in trials involving

the most positively valued stimulus (A), with gamers showing greater value-guided choice behavior in these conditions. This pattern suggests that gaming experience may enhance the consistency with which learned value representations are expressed when generalizing to novel contexts, particularly when high-value options are available. At the same time, the absence of group differences in VLBI indicates that these effects do not reflect a generalized bias toward reward versus punishment, but rather a more precise implementation of learned value information during decision-making (Nussenbaum & Hartley, 2019; Daw, 2011).

These differences in reinforcement learning trajectories may also have implications for individual variability in adaptation across domains. Reinforcement profiles have been linked to clinical traits, with punishment sensitivity associated with anxiety and compulsivity, and reward sensitivity with impulsivity and diminished self-regulation (Bijttebier et al., 2009; Masuyama et al., 2022). While our results did not reveal group differences in valence bias, the greater value-consistent choices in high-value contexts observed in gamers raises important questions about how repeated exposure to reward-rich environments might shape the calibration of motivational systems. Depending on the context, such tuning may promote more efficient learning—as suggested by longitudinal findings showing increased reward responsiveness after gaming training (Lorenz et al., 2015)—or contribute to maladaptive outcomes, as observed in studies on Internet Gaming Disorder reporting heightened sensitivity to reward cues and impaired regulatory control (Weinstein, 2017). More broadly, addiction-related conditions have been associated with alterations in reinforcement learning and in the exploration–exploitation balance, including increased behavioral persistence or reduced exploratory behavior (see Jami et al., 2025 for a review). However, because the present study did not include validated measures of problematic gaming or functional impairment, it cannot determine whether the reinforcement-learning patterns observed here relate to habitual engagement in gaming or to addiction-related mechanisms.

Taken together, these findings indicate that habitual video game play is associated with differences in how learned value representations are expressed during decision-making, rather than in the rate of value updating. Gaming appears to promote more consistent and value-guided choice behavior under conditions of uncertainty, particularly when selecting among competing alternatives with similar expected outcomes. These results are consistent with reinforcement learning accounts in which adaptive behavior depends not only on how values are learned, but also on how efficiently they are translated into action (Sutton & Barto, 2018; Daw & Doya, 2006). Such capacities may hold relevance for both educational design and the understanding of individual variability in adaptive learning.

Limitations

Several limitations should be acknowledged. A plausible explanation for the absence of group differences in the Valence Learning Bias Index (VLBI) during the Transfer phase is that the block configuration always paired the highest-reward option (A) and the lowest-reward option (B) with other alternatives. Although gamers outperformed controls in the Learning and Test phases, particularly under high-uncertainty conditions, controls also discriminated between A and B. As a result, the current structure may have masked any bias in these high-certainty contrasts. Future research could address this by designing transfer blocks that include lower-certainty symbol combinations, where differences in value-guided choices or exploitative choice behavior may emerge more clearly.

Larger samples would also allow exploration of moderating factors such as gaming genre or cumulative playtime. Future studies should examine whether reinforcement learning dynamics and biases differ across video game genres. Specifically, action gamers demonstrate improved probabilistic inference, allowing them to make more accurate decisions based on uncertain or incomplete information (Green et al., 2010). In addition, recent findings indicate that such players exhibit superior "learning to learn" abilities, characterized by a heightened capacity to detect changes in environmental regularities and adjust their value-guided decision policies accordingly (Gao et al., 2024). Strategy games, such as real-time strategy (RTS) games, emphasize planning, multitasking, and dynamic decision-making. These features have been shown to support enhancements in cognitive flexibility, a core aspect of cognitive control (Glass et al., 2013). Similarly, expertise in multiplayer online battle arena (MOBA) games has been associated with superior inhibitory control and greater neural efficiency in frontoparietal networks involved in executive functioning (Ding et al., 2018).

Because our gamer sample included participants from multiple genres (see S1 Table), the observed effects likely reflect reinforcement-relevant design features common across commercial video games—adaptive difficulty that responds to performance, immediate and continuous feedback that supports self-monitoring and rapid policy adjustment, and goal-directed, challenge-rich environments that deliver contingent rewards—rather than mechanics unique to a single genre (Choi et al., 2020). Nonetheless, genre-stratified analyses in future work could test whether these learning benefits are genre-dependent or generalize across

contexts. Controlled, genre-based training interventions would further assess causality and specificity.

Moreover, most research has focused on how video games sustain continuous, goal-contingent rewards. However, many games also involve avoidance learning by imposing strong penalties for failure. Although some experiments have tested how such penalties influence player experience, learning, and interest, their primary focus has been game design rather than cognitive mechanisms (Easterday et al., 2014; Cuerdo et al., 2021). This underexplored feature could contribute to learning biases on the Probabilistic Selection Task (PST). Its impact could be examined through longitudinal training studies that systematically manipulate failure-penalty severity in specific game contexts.

It is also possible that individuals who perform more efficiently under uncertainty also tend to play video games more frequently; this remains speculative. Given the widespread use of video games, individual differences in playing time may not fully account for the effect. Longitudinal or experimental designs are therefore needed to determine whether video game exposure plays a causal role.

Finally, neuroimaging approaches and more advanced computational modeling techniques could further elucidate how experience-driven plasticity in dopaminergic and frontostriatal circuits underpins these behavioral effects. Moreover, intervention studies incorporating controlled gaming paradigms could determine whether structured exposure to game-like reinforcement schedules enhances the efficiency with which learned value representations are translated into action, rather than altering learning rates per se.

Conclusion

In conclusion, our results indicate that habitual video game play is associated with more efficient value-guided decision-making within reinforcement learning contexts, specifically characterized by greater alignment between choices and learned value representations under conditions of high uncertainty, rather than faster trial-by-trial updating. Crucially, these advantages extended partially to the Transfer phase, where gamers exhibited more deterministic and value-consistent choices, although no differences were observed in overall valence bias. This pattern suggests that gaming is associated with enhanced expression of learned values and flexible, state-dependent strategies rather than stable, trait-like motivational biases.

This dissociation highlights the specific nature of the gamer advantage: it is not an optimization of the learning rate itself, but of how learned values are translated into action, particularly under uncertain conditions. Looking forward, these insights offer a tangible framework for applied contexts. In educational gamification and therapeutic design, simply adding points or badges may be insufficient to drive cognitive adaptation. Instead, our findings suggest that interventions should incorporate probabilistic contingencies and noisy feedback loops—key features of commercial games—to effectively engage the neural and cognitive mechanisms that support value-guided decision-making and adaptive behavior under uncertainty.

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Data availability Data and analysis code are available in an OSF repository.

Declarations

Ethical approval This work was approved by the Universidad Autónoma de Baja California (Approval number: 430/2025-1), and was conducted according to the Declaration of Helsinki.

Informed consent Informed consent was obtained from all individual participants included in the present study.

Competing interests The authors declare no competing interests.

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